

REVIEW

CHARACTERIZING AN AUTHOR'S
VOCABULARY

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Summary: This paper gives a brief account of Sichel's (1986) work, and attempts to place it in the general context of stylometric research. We provide a short outline of more recent work carried out on the texts of several French authors, as well as on a poem discovered in 1985, which was attributed to Shakespeare.

1. Introduction

Herbert Sichel was a statistician of very broad interests: one of these was the application of statistics to literary texts, to which he devoted several papers. There is one on word frequency distributions and type-token characteristics published in 1986 in *The Mathematical Scientist*. As a member of the journal's editorial board at the time, I read it with great care and was struck by his grasp of the subject. His death in 1995 has deprived us of a scholarly and versatile colleague; it is fitting that his contributions to the field of linguistic statistics be commemorated.

In his paper, Sichel (1986) was concerned with the type-token relationship $\frac{n_N}{N}$ where n_N is the number of distinct words (types) found in a text consisting of a total of N words (tokens); he was also interested in the frequency of occurrence of particular words in the text. Sichel starts with a probability density function $f(p)$ for the probability $f(p)dp$ of words occurring in a proportion lying between $(p, p + dp)$ in a very long text. In English $0 < p \leq 0.07$, with the proportion $p \sim 0.07$ being that of "the", which is the most frequently used word.

Assuming the generalized inverse Gaussian distribution of the form

$$f(p) = \frac{(2/bc)^g}{2K_g(b)} p^{g-1} \exp(-[p/c] - [b^2c/4p]) \quad 0 < p < \infty, \quad (1.1)$$

where $K_g(b)$ is the Bessel function of the second kind of order g and argument b , Sichel finds

$$n_N \sim [2K_g(b)/bcK_{g+1}(b)][1 - K_g(b(1 + cN)^{\frac{1}{2}})/(1 + cN)^{g/2}K_g(b)]. \quad (1.2)$$

He notes that $g = -\frac{1}{2}$ gives excellent practical results in the majority of cases.

Sichel then studies *hapax legomena* and *hapax dislegomena*, namely words used once only and twice only in a text. He finds formulae for the proportions of these in a text of length N , and observes that for *hapax dislegomena*, with $g = -\frac{1}{2}$, this proportion remains roughly fixed at the level of 15% for a wide range of text length $1,000 \leq N \leq 409,600$. This proves to be in accordance with observation.

He proceeds to consider moments and text length, and concludes with some tests of the stationarity of the word frequency model previously presented. Parameters other than $g = -\frac{1}{2}$ in the distribution $f(p)$ are estimated from data.

Two examples from the works of Macaulay and Dickens are shown to be in good agreement with the model.

Sichel's 1986 paper represents a considerable achievement, giving as it does a mathematical underpinning to several observed characteristics of word frequency distributions and the type-token relationship. To set his results in the context of other studies in linguistic statistics, we shall attempt to give a brief outline of some recent work in the area.

2. Stylometry

Stylometry can be described as the characterization of an author's style by quantitative measures. Among these may be the length of words used by the author; this measure was first considered by de Morgan in 1851, and later used by Mendenhall (1901) to differentiate between the writings of Bacon, Marlowe and Shakespeare. Williams (1970) examined the proportions of words having a length of 1 to 12+ letters in the works of Bacon and Shakespeare, and obtained the results of table 1.

He noted that for Bacon, the largest frequency was that for 3-letter words (22.3%), while for Shakespeare 4-letter words were the most frequent (23.8%). This elementary measure certainly provides a simple test of the difference between the texts of the two authors.

A second possible measure is the length of sentences. Yule (1944) studied several authors, and found that most of Macaulay's sentences contained 11 to 15 words. Coleridge, however, expressed himself mostly in sentences of 21 to 25 words, while Bacon wrote mainly in sentences of 31 to 35 words. Williams (1970) was able to show that sentence length for a variety of authors followed

Table 1. Frequencies of words of 1 to 12+ letters in Bacon and Shakespeare.

Word length	1	2	3	4	5	6
Bacon %	2.5	19.8	22.3	17.5	10.0	7.5
Shakespeare %	4.8	17.6	22.5	23.8	12.4	7.1

7	8	9	10	11	12	>12
7.0	4.5	3.7	2.5	1.2	0.5	1.0
5.3	3.1	1.8	0.9	0.3	0.2	0.2

a log-normal distribution, and was able to distinguish the distribution parameters for Chesterton, Shaw and Wells.

The most characteristic measure of an author's style is his or her vocabulary. As readers, all of us recognize an author's use of vocabulary in a particular text without relying on any quantitative measure. Jane Austen's witty conversations and James Joyce's stream of consciousness passages are readily recognizable. But as statisticians, we prefer to derive several quantitative stylometric measures from a text, if only to support our intuition.

The first to discuss the frequency of words in a literary text was Zipf (1932). He enunciated Zipf's law, which states that if k different words in a text are ranked in the order $r = 1, 2, \dots, k$ of their frequencies, with $r = 1$ corresponding to the word with the highest frequency, then the frequency of the r -ranked word is given by

$$f(r) = Ar^{-B} \quad r = 1, \dots, k, \quad (2.1)$$

where usually $B \geq 1$, and $A = \left[\sum_{r=1}^k r^{-B} \right]^{-1}$. While this fits the data quite well, Sichel (1975) has shown that a compound Poisson model gives slightly better results in English as well as in other languages.

The frequencies of particular words may be used to establish the authorship of a given text; this is what Mosteller and Wallace (1964) did in their book on *The Federalist* papers. Both Alexander Hamilton and James Madison wrote for *The Federalist* during the American Revolution; of the 69 essays in that publication attributed to them, it was known that 43 were written by Hamilton and 14 by Madison, but another 12 could not be assigned with any certainty to either one or other of them. Mosteller and Wallace concentrated on the frequency of certain words used by the authors: they noted that Hamilton wrote "while" where Madison used "whilst" and that Hamilton favoured the word "upon" 18 times more frequently than Madison. By analyzing their data by four different methods, two of them Bayesian, one based on the rate of use of key words, and another on a linear discrimination procedure, they came to the conclusion that the 12 unattributed articles had been written by Madison.

Yet another stylometric measure is the sheer size or richness of an author's vocabulary. When a text is written, the total number N of words in it (tokens) will contain n_N distinct words (types), and the relation between n_N and N is referred to as the type-token relationship. Since an author's types are selected from an active vocabulary which is usually assumed to be of some fixed size M , one would expect n_N to approach M asymptotically. One of the problems of stylometry is to estimate M from the number of types in a text, and for this, one requires some acceptable type-token model. One such is that of Sichel (1986), resulting in equation (1.2); there, $M = 2K_g(b)/bc K_{g+1}(b)$ is the asymptote of n_N , which is approached very slowly, with $g = -\frac{1}{2}$ usually in practice. We now present alternative type-token models.

3. Some type-token models

One of the most elementary type-token models assumes that the number of types $\{n_N; N \geq 1\}$ is a simple Markov chain with M states, and transition probabilities of the form

$$p_{jj} = j/M, \quad p_{jj+1} = 1 - j/M = (M - j)/M, \quad j = 1, \dots, M. \quad (3.1)$$

This indicates that if for some number N of words in a text, there are j types, then on adding an extra word, the probability that it is a new type is $(M - j)/M$ so that this type is chosen at random from among the $M - j$ not previously used. Clearly this is not quite true, but it is acceptable as a first approximation. A refinement of this model can be obtained by dividing the author's active vocabulary into two main groups, Group I consisting of nouns, adjectives, verbs and adverbs, and Group II of pronouns, prepositions, conjunctions, auxiliary verbs, articles and interjections. A typical dictionary will contain around 100,000 words of the first group and 500 of the second. Using this and other similar models, Gani and Saunders (1976) and Efron and Thisted (1976) estimated that Shakespeare's vocabulary might well have exceeded 50,000 words; an educated person usually manages with 15 to 20,000.

A somewhat more complicated model was studied by Daley, Gani and Ratkowsky (1991) in an effort to improve the agreement between theory and practice. Detailed data was available for several French authors, among them Corneille, Racine, Constant, Giraudoux, Proust and Zola which could be used to verify the goodness of fit of the type-token relationship. A Markov chain $\{n_N; N \geq 1\}$, with M states as before, was considered for which the transition

probabilities were now

$$p_{jj} = 1 - p_{jj+1} p_{jj+1} = d(1 - j/M)^k, 1 \geq d > 0, k > 1. \quad (3.2)$$

We see that the new $p_{jj+1} < (1 - j/M)$ of (3.1), and this inequality becomes more pronounced as j approaches M , reflecting the fact that authors may find it difficult to find a new type as they approach the upper limit of their vocabulary. One interesting result emerging from the study was that even for very large values of N , one did not seem to reach the limit M . This led to a variant of the model (3.2) in which both k and M were allowed to increase indefinitely in such a way that $k/M \rightarrow K$ as $M \rightarrow \infty$. In this particular model, the expectation of n_N was found to be

$$E(n_N) = K^{-1} \ln(1 + NdK). \quad (3.3)$$

Numerical results for the estimates of the parameters $\hat{d}$ and $\hat{K}$ from the data sets for six French authors, using a least squares fit are given in table 2.

Table 2. Parameter estimates fitted to model (3.3) by least squares.

Author	N	n_N	$\hat{d}$	$10^3 \hat{K}$
Constant	27,660	2,509	0.446	1.062
Racine	158,899	2,931	0.203	1.381
Corneille	532,800	4,606	0.326	1.121
Giraudoux	671,364	15,771	0.211	0.218
Proust	1,267,069	18,322	0.135	0.193
Zola	2,874,755	19,337	0.091	0.211

A plot of $\hat{d}$ and $\hat{K}$ on coordinate axes indicates the existence of two clusters: the first three authors are near the origin and the last three some distance from it. The value of $\hat{K}$ appears to be a direct measure of the size of the author's

“besots”, “exiles”, “inflection”, “joying”, “scanty”, “speck”, “tormentor”, and “explain”. On the basis of Shakespeare’s known works, it is possible to predict the expected numbers of types occurring 0, 1, 2,..., 99 times respectively in a poem of 329 words; these are given in table 3.

Table 3. Expected and observed counts of words appearing x times in a poem of 429 words.

x	0	1	2	3	4	5
Expected count	6.97	4.21	3.33	2.84	2.53	2.43
Observed count	9	7	5	4	4	2

10	30	50	70	90	95	99
1.62	0.96	0.68	0.49	0.37	0.34	0.32
1	4	0	0	0	1	0

In order to add breadth to their analysis, they also considered poems by Ben Jonson, Christopher Marlowe, and John Donne as well as four known Shakespeare poems from *Cymbeline*, *A Midsummer Night's Dream*, *The Phoenix and the Turtle* and *Sonnets 12-15*. Three different tests were used, all relying on a regression model in which the observed counts follow the Poisson distribution with a mean

$$m_x = e_x \exp(b)[x + 1]^c,$$

where e_x is the expected count for words appearing x times. Of the three tests, that which proved to be the best Shakespeare discriminant was the third, namely that testing for $c = 0$. This appeared to show that there was no appreciable semantic difference between the new and the four known poems. Despite

this conclusion, there remains serious doubt as to the authorship of "Shall I die?"

5. Concluding remarks

This brief survey should indicate that there remain many problems of stylometry which await solution. Sichel's contributions to the field have been pioneering efforts of great value. However, the example of the poem attributed by Taylor to Shakespeare, with the lingering uncertainty about its authorship despite Thisted and Efron's stylometric analysis, point to the need for more sophisticated stylistic measures than those already available. A survey by Holmes (1994) on authorship attribution provides a recent account of work in this area. If the present paper stimulates further research in stylometry among statisticians, it will have achieved its purpose.

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